Fundamentals of Electrical and Electronics Engineering (FEEE)

SUB CODE: ECE1002

T-P-C: 3-2-4

Lecture-9

Faculty

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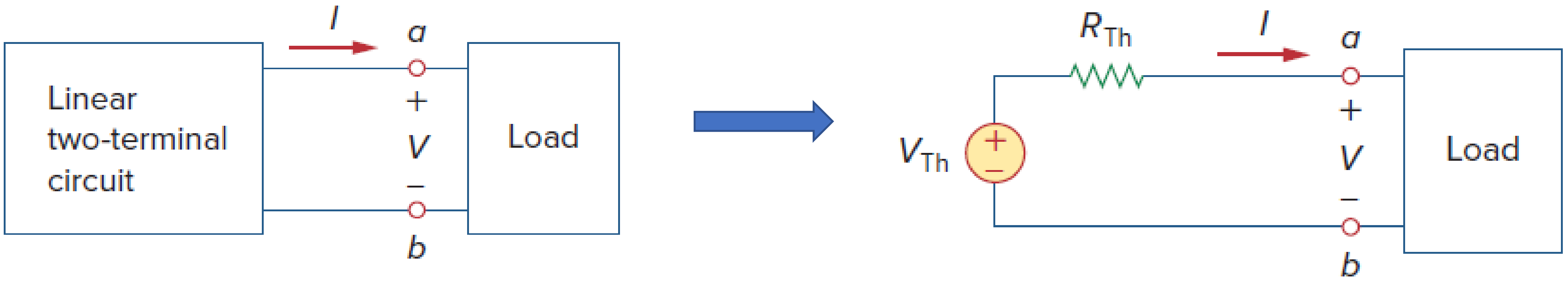
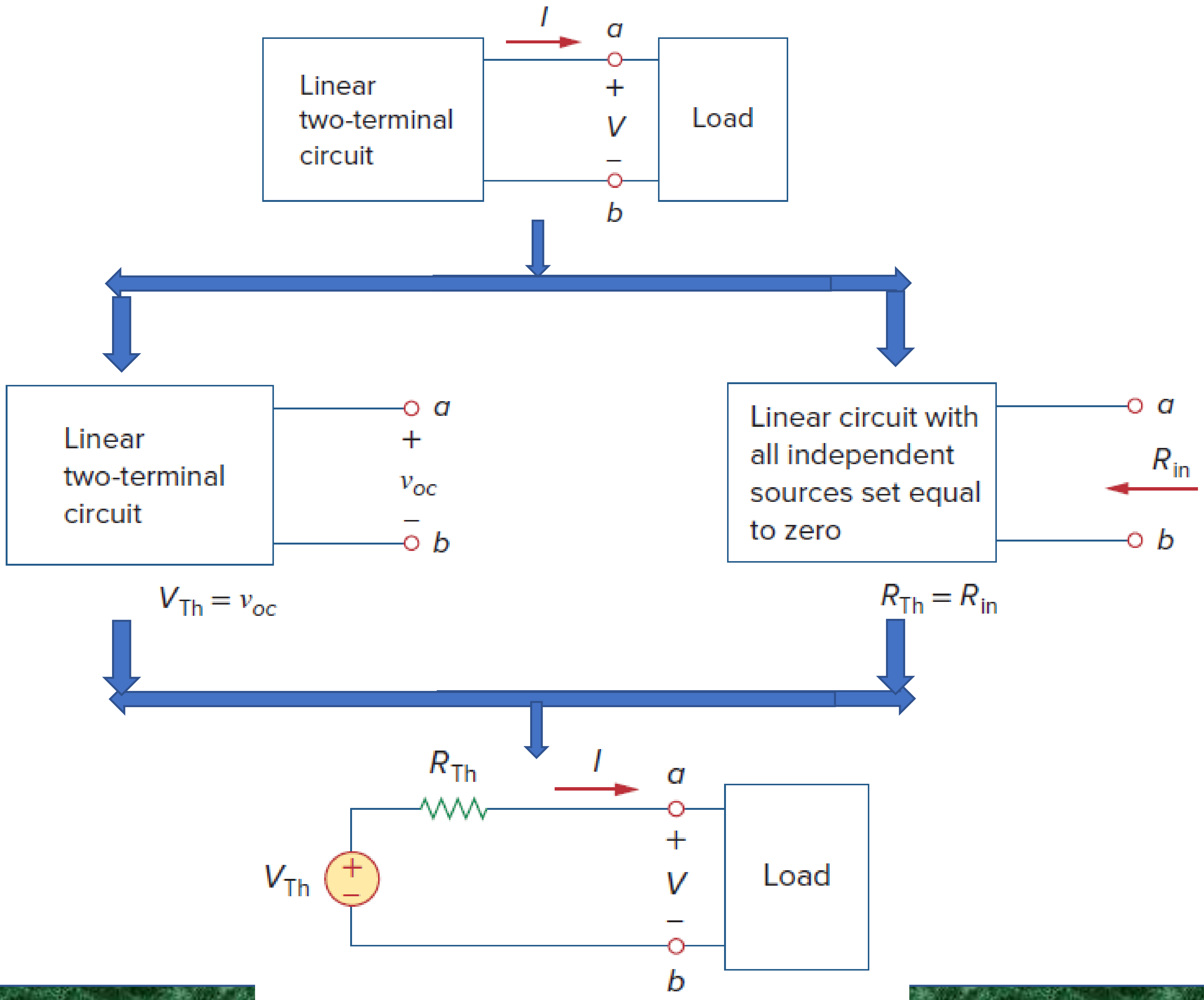
Network Theorems

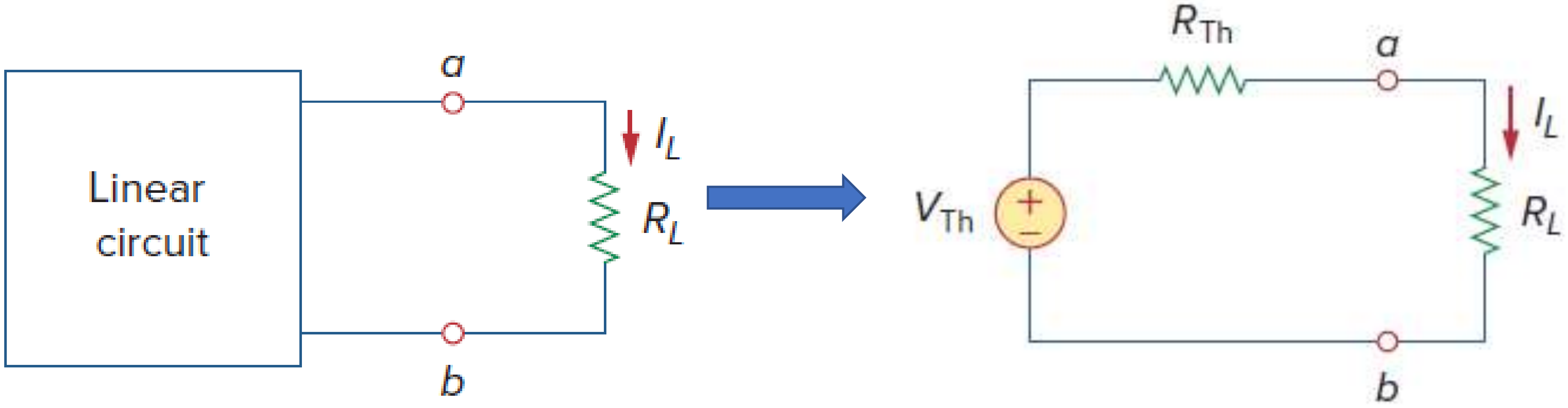
1. Superposition Theorem
2. Thevenin Theorem
3. Maximum Power Transfer Theorem

2.Thevenin's Theorem

- Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th}.
- V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

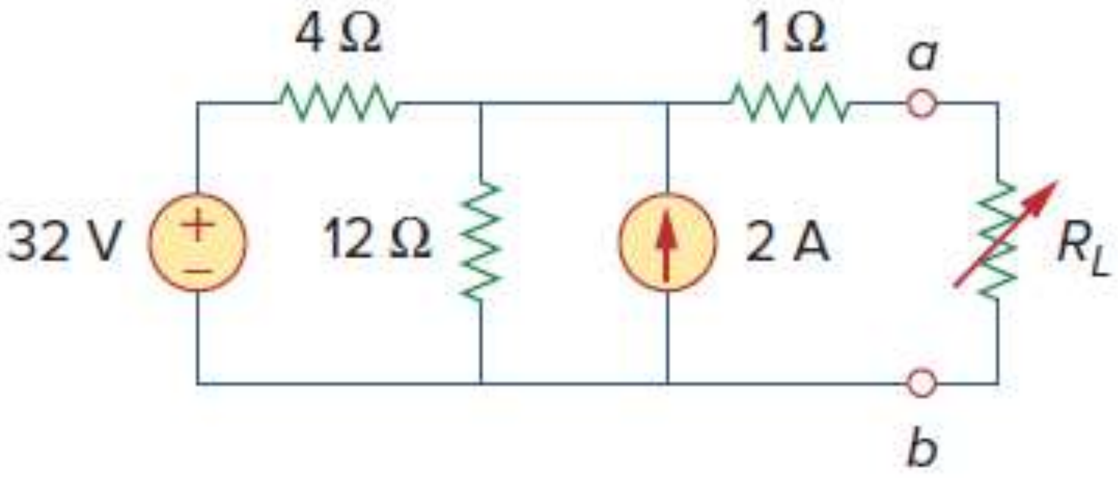



I
a
Linear
two-terminal
circuit
+
V
−
Load
b
Linear
two-terminal
circuit
a
+
v_{oc}
−
b
Linear circuit with
all independent
sources set equal
to zero
a
R_{in}
b
$V_{Th} = v_{oc}$
$R_{Th} = R_{in}$
R_{Th}
I
a
V_{Th}
+
V
−
Load
b

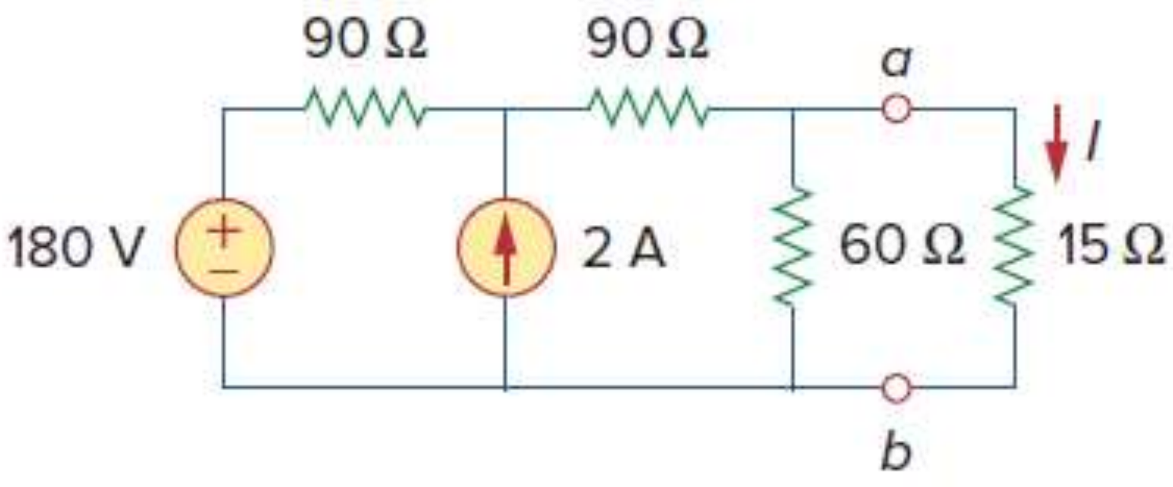

Linear circuit
a
b
I_L
R_L
R_{Th}
V_{Th}
+
−

P-1

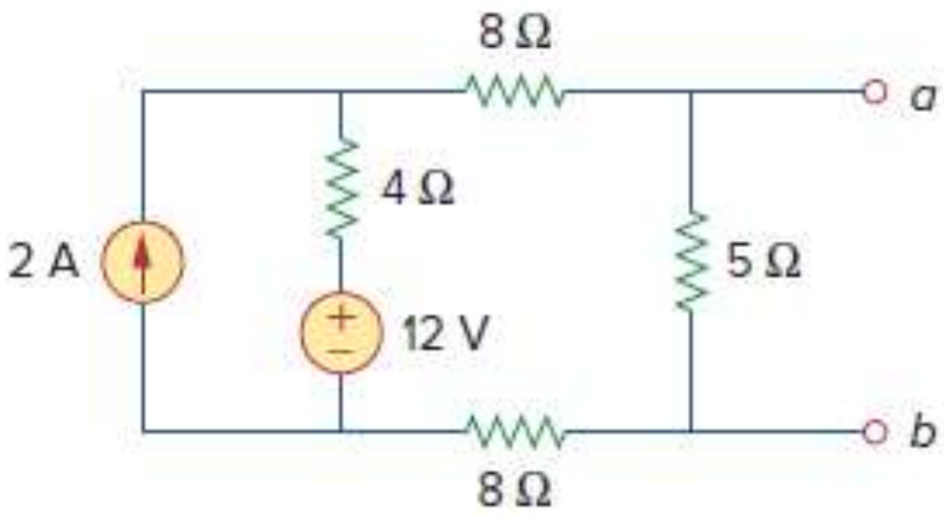
Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals *a*-*b*. Then find the current through R_L = 10 Ω, 15 Ω, and 40 Ω.



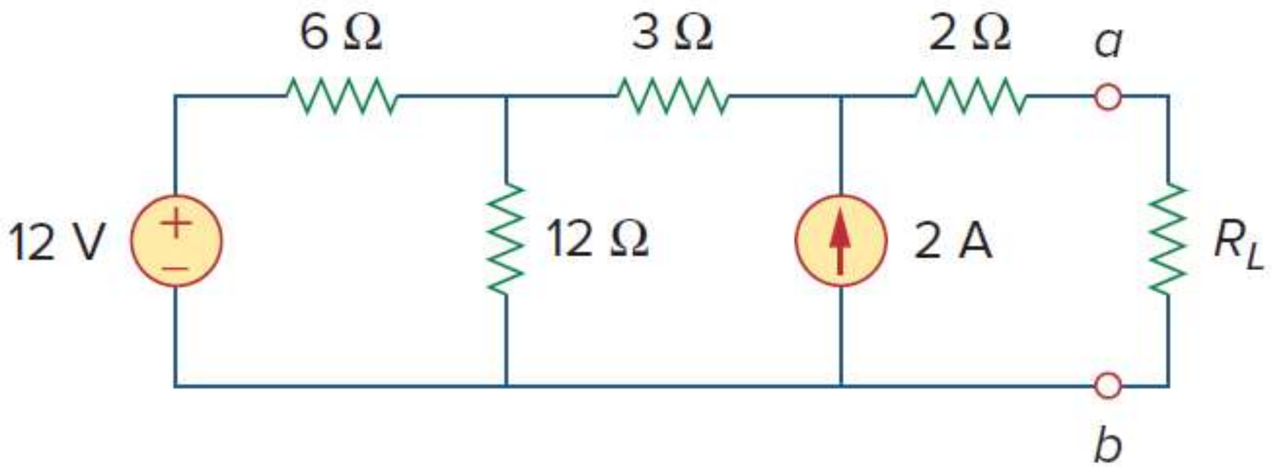
P-2: Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals *a*-*b*. Then find the current through 15 Ω.



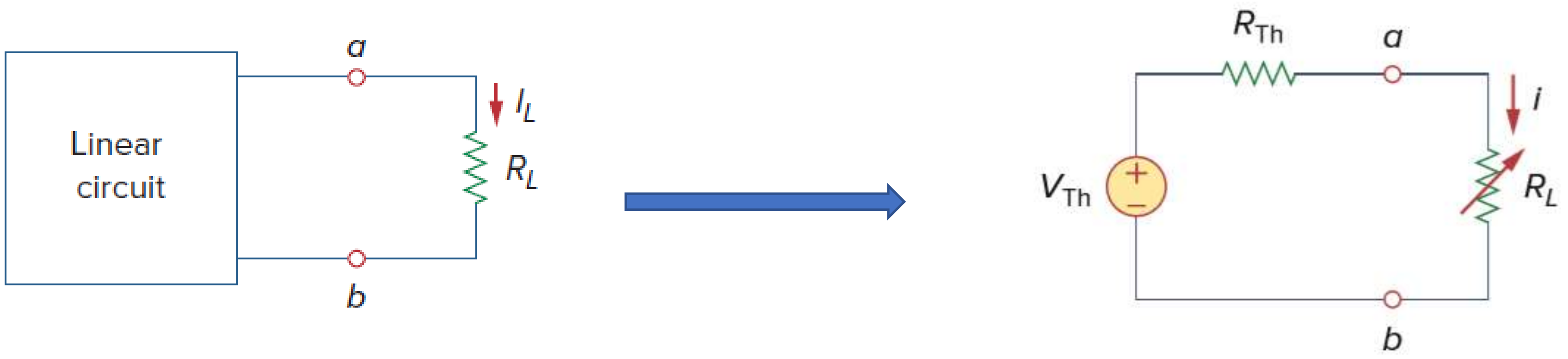
P-3: Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals *a-b*. Then find the current through if R_L=10 Ω between a-b.

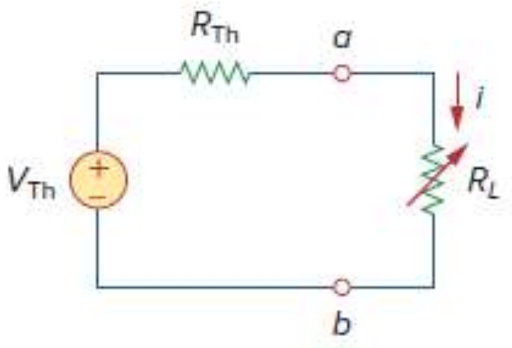


P-4: Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals *a*-*b*. Then find the current through R_L=10 Ω.



III). Maximum Power Transfer Theorem





$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L}\right)^2 R_L$$

To prove the maximum power transfer theorem, we differentiate p with respect to R_L and set the result equal to zero. We obtain

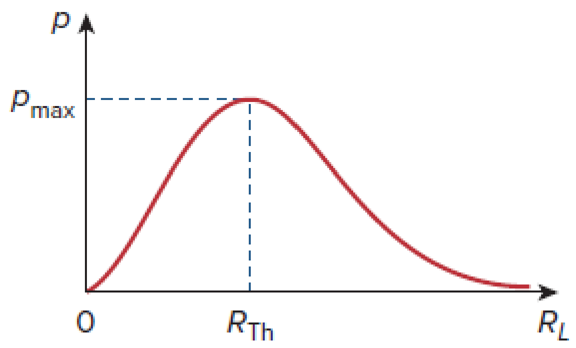
$$\frac{dp}{dR_L} = V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4}\right]$$

$$= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3}\right] = 0$$

$$0 = (R_{Th} + R_L - 2R_L) = (R_{Th} - R_L)$$

The maximum power transferred to the load when

$$\boxed{R_L = R_{Th}}$$

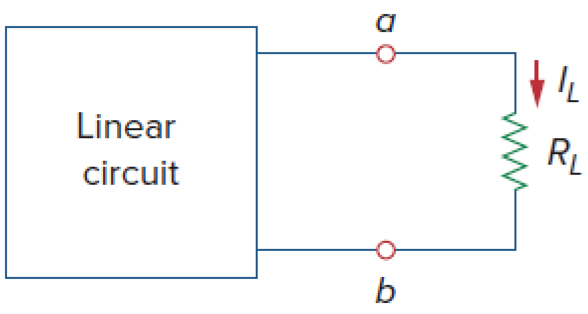


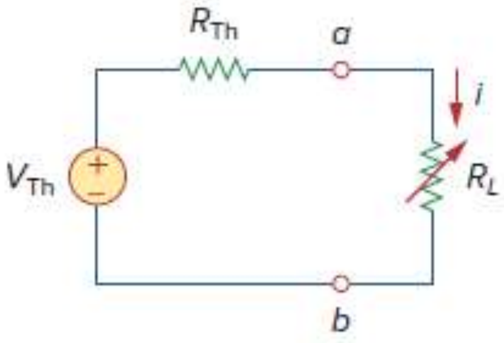
$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L}\right)^2 R_L$$

The maximum power transferred to the load is

$$\boxed{p_{max} = \frac{V_{Th}^2}{4R_{Th}}}$$

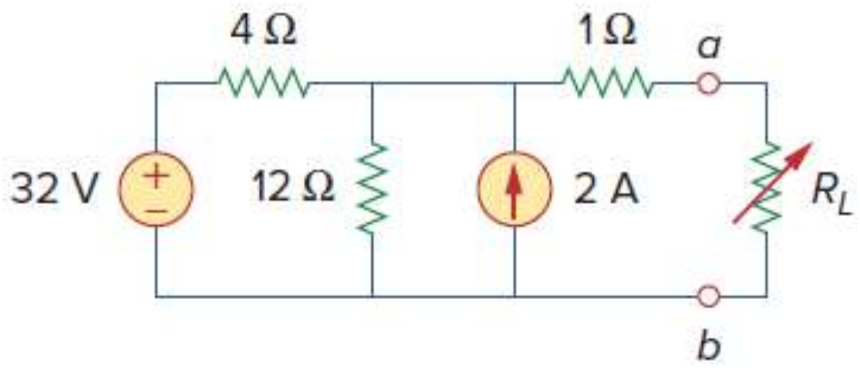
Maximum Power Transfer Theorem





- Maximum Power transferred to the load when load resistance equal to source resistance.

Find the value of R_L for maximum power transfer in the circuit of Fig. Find the maximum power transferred to the load.



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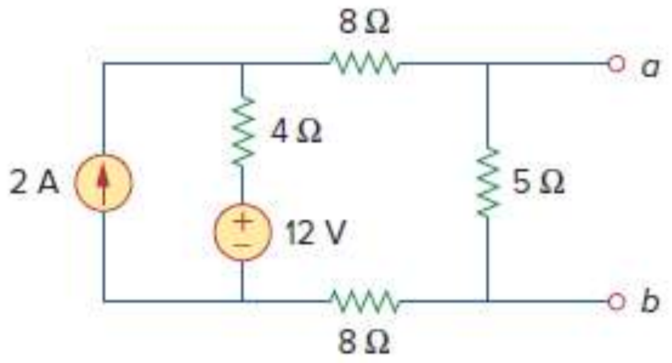

Find the value of R_L for maximum power transfer in the circuit of Fig. Find the maximum power transferred to the load.